

HTML

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HTML = HyperText Markup Language

HTML is the skeleton structure of every webpage you have ever visited. Before YouTube videos play, before Instagram posts, before Google displays results — that browser first renders and parses HTML. It is not a programming language. It is a **markup language** — meaning it uses predefined tags to describe content and define its structure and semantic meaning.

HYPERTEXT

- “Hyper” means beyond normal.
- Text is what a text that can link to other stuff.
- This is what `<a href="https://www.youtube.com/watch?v=...">` does in the 1990s, this was revolutionary.

Markup Language

- Markup = applying labels to content. Like a teacher marking exam paper with red pens. You mark up text content with tags to tell the browser what each piece means.

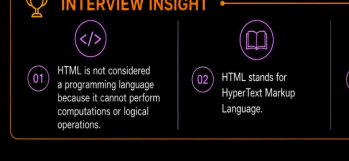
RELATE WITH REAL-WORLD

HTML is blueprint ki तरह hoga hai, jaise ki architect plan ka structure design karta hai.



In early HTML, webpage is structure our content by using tags like this:

HTML vs CSS vs JS

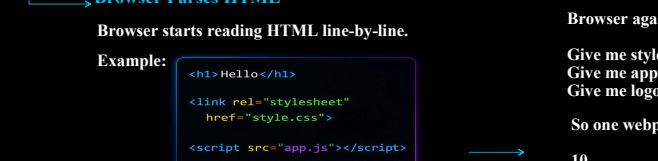


Therefore, HTML provides the structure, CSS handles presentation, and JavaScript adds behavior.



Client vs Server

Every website interaction is a conversation between two computers: the Client (your device) and the Server (a computer somewhere in the world that stores files).



Browser mostly receives HTML, CSS, JS, Images, Fonts, Video/Audio.

Browser parses HTML.

Browser starts reading HTML line-by-line.

Example:

```
<html><head></head><body><h1>Hello World!</h1></body></html>
```

New browser discovers:

- JS needed
- Images needed

Then browser sends MORE requests.

Additional Requests Happen

Browser again asks server:

- Give me styles.css
- Give me script.js
- Give me img.jpg

So our webpage now create:

- JS
- img
- styles

Relate with Real-World:

Client = Customer at a restaurant. Server = Kitchen. You (client) place an order (request). The kitchen (server) prepares the food and sends it back (response). You don't leave HDN they made it — you just receive the result.

HTML = human built language, CSS = style design, colour how looks things person ka colour of dress.

JS = human built, use your logic to do a human like to different task things.

Why HTML was Created

- The 1980s Problem:** Researchers wanted a simple, easy-to-use language for creating documents. Existing languages like TeX and Troff were too complex and focused on typesetting.
- The Solution: Hypertext:** The idea was to create a language that could describe the structure and content of documents, and also allow for links between different parts of the document.
- The Impact:** Hypertext made it easy to connect, organize, and share information.

1989 — WWW idea proposed

1990 — HTML created

1991 — Mosaic browser became popular

Important: HTML had did not exist the Internet.

HTML was the Internet's anatomy and core-friendly.

The WWW website made for World Wide Web — a system of interconnected documents linked using hypertext.

Internet vs WWW

Internet: The physical/global network of computers.

WWW (World Wide Web): A system of interconnected webpages running ON the Internet.

Internet already existed

WWW made it user-friendly and accessible

That's why HTML and hypertext became revolutionary.

First Website Ever Created

Created by Tim Berners-Lee at CERN.

It was launched in 1990.

It was one of the world's largest scientific research organizations.

CERN = European Organization for Nuclear Research

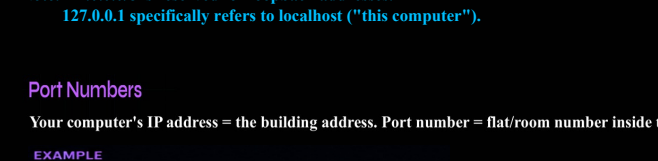
Link: <https://info.cern.ch/journeys/WWW/TheProject.html>

LOCALHOST & PORT NUMBERS

On Your Local Machine

When you create a website, you need a server to serve the files. But do you need a big server in another country? Not Your own computer can act as a server for testing.

localhost — your own computer acting as a server. You access both the client AND the server — open computer to a local server has jaisa hai.

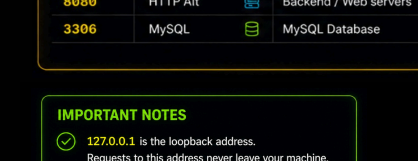


Note: 127.0.0.1 is reserved for localhost address.

127.0.0.1 is specifically refers to localhost ("this computer").

Port Numbers

Your computer is IP address — the building address. Port number — door number inside the building.



PORT	SERVICE	USE
80	HTTP	Web browser (default port)
443	HTTPS	Secure websites
8080	Default HTTP	Proxy / Reverse proxy servers
3000	Local Server	VS Code Live Server
8000	HTTP API	Backend / Web services
5000	HTTP API	Backend / Web services

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IMPORTANT NOTES
• 127.0.0.1 is the standard address.
• Always use 127.0.0.1 when testing your website.
• 127.0.0.1 is a loopback address that routes traffic back to the same machine.
• You can use other IP addresses on the same machine, but 127.0.0.1 is the most common.
• Always use 127.0.0.1 when testing your website.

INTERVIEW INSIGHT
• 127.0.0.1 is used as the default address for localhost.
• Port 80 is the default port for HTTP.
• Port 443 is the default port for HTTPS.
• Always use 127.0.0.1 when testing your website.
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HTML Tags, Elements & Syntax

HTML is a TAG

Tags are keywords used to tell the browser how to display the content. Tags are the building blocks of HTML. Every tag has a purpose and sets the format. Your browser interprets tags like this:



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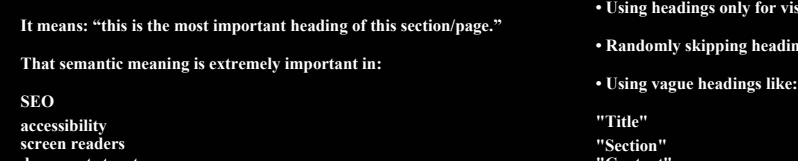
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Now it's time to implement these concepts practically.

Headings h1 - h6

Headings are used to structure the content of a webpage. They are the building blocks of HTML. Every tag has a purpose and sets the format. Your browser interprets tags like this:



HEADINGS	DESCRIPTION	EXAMPLE
h1	Main Page Title	h1>Main Page Title</h1>
h2	Section Header	h2>Section Header</h2>
h3	Sub-section Header	h3>Sub-section Header</h3>
h4	Section Header	h4>Section Header</h4>
h5	Section Header	h5>Section Header</h5>
h6	Section Header	h6>Section Header</h6>

SEO Basics of Headings

Search engines and website technologies analyze HTML headings to understand:

- Page structure
- Content hierarchy
- Important topics

Proper heading usage improves:

- Readability
- Accessibility
- Content organization

Important Interview Insight

h1 does NOT mean "make text bigger"

It means "this is the most important heading of this section/page."

That semantic meaning is extremely important in:

- SEO
- Accessibility
- Screen readers
- Document structure

Best Practices

- Prefer one primary h1 for the main page topic.
- Use heading hierarchy: h1 → h2 → h3 → h4 → h5 → h6
- Use headings to describe section content clearly.
- Maintain logical heading structure where possible.
- Use CSS for styling — not heading tags purely for font size.
- Include important keywords naturally in headings.
- Ensure headings improve accessibility and screen-reader navigation.

Common Mistakes

- Using headings only for visual styling.
- Randomly skipping heading hierarchy without structural reason.
- Using vague headings like:

- "Title"
- "Section"
- "Content"

- Using multiple unrelated h1/h2 tags without clear structure.
- Using heading tags where semantic meaning is unnecessary.

Text Tags: <p>, , , <code>

Text Tags	DESCRIPTION	EXAMPLE
<p>	Paragraph	<p>This is a paragraph of text. It is used to group multiple lines of text together. It is the most basic text tag in HTML.
	Strong	This text is important. This tag is used to emphasize text. It is often used for headings and sub-headings.
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